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Fixation and saccade classification with the I-VT filter

This article describes how iMotions classifies fixations with the I-VT Filter, which is the default alternative to the [Duration dispersion filter](#) as well as a [classification based on Hidden Markov Models](#). I-VT filters are typically used for higher-sampling eye tracking data (200 Hz or more). For a scientific overview, please see [Holmqvist et al. \(2011\) pp.147-175](#).

About the I-VT fixation classification filter

What is the I-VT filter?

I-VT is an abbreviation for velocity-threshold identification. It is used to analyse the raw eye tracking data and classify fixations and saccades by comparing the speed of the eye's movements to a velocity threshold. If the eye is moving slower than this threshold, the I-VT filter classifies a fixation. If the eye is moving faster than this threshold, the I-VT filter classifies a saccade. It can happen that a classification is not possible, for example because of missing data or because additional parameters (such as duration thresholds) are violated. In these cases, the I-VT filter classifies "other". Please note that other types of eye movements such as smooth pursuit cannot be classified with an I-VT filter; they get classified as either a fixation or a saccade, depending on the set velocity threshold.

Basic principles of eye movement classification with the I-VT filter (video):

Computation in detail:

- Data preparation:
 - From the binocular gaze data in x- and y coordinates for the left and right eye, an average gaze point is computed. If data is missing from one eye, it is replaced with the gaze coordinate from the other eye.
 - If gaze or distance data is missing from both eyes, the data gap can be filled with a linear interpolation from the start to the end of

the gap. The maximum length for this interpolation can be modified with the parameter *Max Gap Length* in the section *Gap Fill-in (interpolation)*. The interpolation can also be turned off.

- To smooth out noise in the data, it is possible to apply either a *Moving Average* filter or a *Median* filter inside a moving window around the current sample. The size of this window can be modified with the *Window Size* parameter in the section *Noise Reduction*. This smoothing is optional.
- Signal processing:
 - To compute the gaze velocity at a given sample, the algorithm computes the angular velocity between the first and last sample inside a time window around the current sample. The size of the time window is defined by the *Window Length* parameter in the *Fixation Filter Parameters* section. For this, the distance between the first and last sample inside this time window (in pixels, obtained from their x- and y gaze coordinates) is converted to an angle by taking the respondent's distance to screen into account (as measured by the eye tracker). Dividing this angle by the time lapse between the two samples gives a velocity value in °/s, the velocity computed for the current sample. If a sample inside a time window are missing, no velocity is computed.
 - Gaze acceleration is computed based on the difference in velocity between the current and its previous sample.
 - For every sample, its velocity is compared against the threshold defined by the *Velocity Threshold* parameter in the *Fixation Filter Parameters* section. If the velocity is equal or higher, the sample is classified as belonging to a saccade. If it is lower, the sample is classified as belonging to a fixation.
- Corrective steps:
 - Noise in the data can lead to jumps between the gaze points of a fixation, which lead to the classification of a saccade and therefore interruption of the fixation. To correct this, fixations in close temporal or spatial neighborhood can be merged together using the *Max Time Between Fixations* and *Max Angle Between Fixation* parameters in the *Merge Adjacent Fixations* section.
 - Short fixations can be discarded using the *Minimum Fixation Duration* parameter in the *Discard Short Fixations* section. This can be desired because too short fixations are sometimes

regarded as not meaningful when used as an indicator of cognitive processing. Applying this option will compare every classified fixation against the *Minimum Fixation Duration* parameter and classify it as an unknown event in case the fixation is shorter than this threshold.

- Computation of fixation metrics:
 - Fixations are numbered from start of the recording to its end, as well as from start of the stimulus to its end. Only fixations with their start and end on the same stimulus get counted as belonging to this stimulus.
 - The start time of the fixation is calculated as the average between the timestamp of the first sample of the fixation and the sample preceding it. The end time of the fixation is the average of the timestamps of the last sample of the fixation and the one succeeding it. The time difference between fixations start and end times is equal to the fixation's duration.
 - The x- and y coordinates of the fixation's centroid are determined by finding the point with the closest distance to all samples of the fixation.
 - Dispersion of the fixation is calculated with a root mean square of the samples belonging to it.
- Computation of saccade metrics:
 - Saccades are numbered from start of the recording to its end, as well as from start of the stimulus to its end. Only saccades with their start and end on the same stimulus get counted as belonging to this stimulus.
 - The start time of the saccade is calculated as the average between the timestamp of the first sample of the saccade and the sample preceding it. The end time of the saccade is the average of the timestamps of the last sample of the saccade and the one succeeding it. The time difference between saccade's start and end times is equal to the saccade's duration.
 - The gaze locations of the first and last sample belonging to the same saccade are used to compute the amplitude of the saccade.
 - Velocities of all samples belonging to the same saccade are compared, and the highest one is classified as the saccade's peak velocity.
 - Accelerations of all samples belonging to the same saccade are compared, and the highest (positive) value is classified as the

saccade's peak acceleration.

- Accelerations of all samples belonging to the same saccade are compared, and the lowest (negative) value is classified as the saccade's peak deceleration.

Settings and export options in iMotions

Parameters in the "Gaze Analysis" tab at study level:

The screenshot displays the 'Gaze Analysis' settings tab in the iMotions software. The top navigation bar includes tabs for STUDY SETTINGS, SENSORS, GAZE ANALYSIS (highlighted), LIVE MARKERS, EXPORT, RECORDED DATA, and LATENCY SETTINGS. The settings are organized into several sections:

- Fixation Filter:** A dropdown menu set to 'I-VT Fixation'.
- Gap Fill-in (Interpolation):** A checkbox that is unchecked. Below it, 'Max Gap Length' is set to 75 ms, with a 'Reset' button.
- Noise Reduction:** A checkbox that is unchecked. Below it, a dropdown menu is set to 'Moving Average', 'Window Size' is set to 3 samples, and there is a 'Reset' button.
- Fixation Filter Parameters:** A section with two rows of settings, each with a 'Reset' button.
 - Row 1: 'Window Length' is 20 ms, 'Velocity Threshold' is 30 degrees/second.
 - Row 2: 'Discard Short Fixations' checkbox is unchecked, 'Minimum Fixation Duration' is 60 ms.
 - Row 3: 'Merge Adjacent Fixations' checkbox is unchecked, 'Max Time Between Fixations' is 75 ms, 'Max Angle Between Fixation' is 0.5 degrees.

At the bottom right, there are 'Undo' and 'Update' buttons.

- **Fixation Filter:** Option to choose between the I-VT filter explained in this article and the [Duration Dispersion filter](#).
- **Gap Fill-in (interpolation):** If selected, missing data is linearly interpolated from the last recorded sample before the data gap until the first recorded sample after the data gap, given the gap is no longer than the indicated maximum gap length. Unless interpolated, samples with missing data will be classified as an unknown eye movement.
- **Noise Reduction:** Smoothes the signal and replaces the coordinates of the current sample with a newly computed value. Please bear in mind that when the noise reduction is applied on data from a low-sampling eye tracker, one risks to filter out the signal.
 - Users can choose between a *moving average* or a *median* filter applied on the samples inside the window. The moving average has a stronger smoothing effect than the median filter.
 - **Window Size:** Determines across how many samples the smoothing shall take place. The more samples are included, the

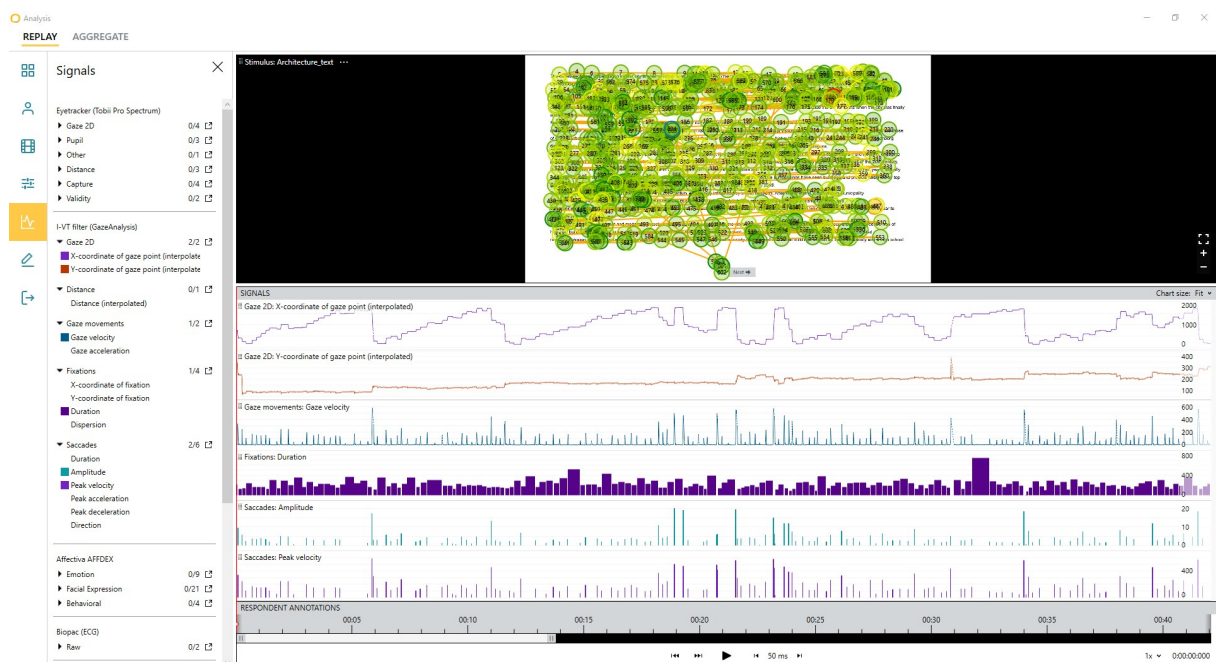
stronger the smoothing.

- **Fixation Filter Parameters**

- **Window Length:** Size of the time window around the current sample, used to compute the current gaze velocity. Smaller time windows will make the filter more sensitive to rapid changes in the gaze velocity. If the window size is too small, the two neighboring samples are used even if they fall outside the time window.
- **Velocity Threshold:** Threshold against which the sample's velocities are compared to classify them as parts of a saccade or fixation
- **Discard Short Fixations:** Fixations falling below this threshold will be reclassified as an unknown eye movement.
- **Merge Adjacent Fixations:** Fixations in a close temporal or spatial neighborhood can be merged together

Results:

The result of the fixation classification will be visible in the data replay.



Depending on the visualization you chose, some parts of the graphs will be shown as a dotted line because there is no data available for this time point (for example if you are plotting the amplitude of a saccade, the time while a fixation was happening will be shown as a dotted line). Therefore, we recommend using the "Area" visualization for Fixation and Saccade visualizations.

BUILT IN PROCESSING

R PROCESSING

Export

AOI

☒ AOI Metrics Export

☐ Affective Circumplex
 ☐ Descriptive Statistics Export

ECG

☐ ECG Metrics Export
 ☐ EEG Frontal Asymmetry Metrics Export

Eye Tracking

☐ Eyetracker Blink Detection Metrics Export

Facial Expressions

☐ AFFDEX Blink Detection Metrics Export
 ☐ AFFDEX Metrics Export
 ☐ FACET Metrics Export
 ☐ RealEyes Metrics Export

GSR

☐ GSR Metrics Export

Settings

Please select export type(s):

Threshold for dwell classification [ms]

Amount of first fixations to exclude

Exclude fixations starting before AOI onset or ending after AOI offset

Include Annotation markers in the fixation & saccade tables

Include Scene markers in the fixation & saccade tables

Individual AOI metrics;Aggr...

☒ (Select All)
 ☒ Individual AOI metrics
 ☒ Aggregated AOI metrics
 ☒ Fixation & saccade tables

OK Cancel

Description

Produces files export for AOI metrics.

More information about this notebook and its parameters can be found in the [Help Center](#)

Replay

Aggregate

AOIs

Gaze Map

Heatmap

Survey

Signal Processing

Export

Export
 Help

Sensor data

☒ Eyetracker Tobii Pro Spectrum (0/15)

R Analysis GazeAnalysis I-VT filter (25/25)

Gaze ZD
Select none

- ☒ X-coordinate of gaze point (interpolated)
Gaze data X-coordinate (relative to top-left corner of the screen). Average of the X-coordinates of the left and right eyes.
- ☒ Y-coordinate of gaze point (interpolated)
Gaze data Y-coordinate (relative to top-left corner of the screen). Average of the Y-coordinates of the left and right eyes.
- ☒ X-coordinate of gaze point (interpolated)
Gaze data X-coordinate (relative to top-left corner of the screen) with missing coordinates interpolated. Average of the X-coordinates of the left and right eyes.

Distance
Select all Select none

- ☒ Distance (interpolated)
Estimated distance between the eye-tracker and the eyes, with missing values interpolated. Average of the eye tracker's distance to the left and right eye.

Gaze movements
Select none

- ☒ Gaze velocity
Angular velocity of the gaze at the current sample point. i.e. how fast the eyes are moving at this point in time.
- ☒ Gaze acceleration
Angular acceleration of the gaze at the current sample point. i.e. how much the eyes increased in speed at the current sample.

Fixations
Select none

- ☒ Index
Fixation number, counting from start of the recording.
- ☒ Index (per stimulus)
Fixation number, counting from start of the stimulus.
- ☒ X-coordinate of fixation
X-coordinate of the fixation centred (relative to top-left corner of the screen).
- ☒ Y-coordinate of fixation
Y-coordinate of the fixation centred (relative to top-left corner of the screen).

Saccades
Select none

- ☒ Index
Saccade number, counting from start of the recording.
- ☒ Index (per stimulus)
Saccade number, counting from start of the stimulus.
- ☒ Start time
Start time of the saccade (relative to recording start).
- ☒ End time
End time of the saccade (relative to recording start).
- ☒ Duration
Duration of the saccade.

☒ Affective AFFDEX (0/35)

☒ Affective AFFDEX Landmarks (0/105)

DATA PREVIEW (9)														
ID	TimeStamp	EventSource (SlideEvents)	SlideEvent (SlideEvents)	StimType (SlideEvents)	Duration (SlideEvents)	CollectionPhase	SourceStimulusName (SlideEvents)	EventSource (R Analysis GazeAnalysis I-VT filter)	Gaze X (R Analysis GazeAnalysis I-VT filter)	Gaze Y (R Analysis GazeAnalysis I-VT filter)	Interpolated Gaze X (R Analysis GazeAnalysis I-VT filter)	Interpolated Gaze Y (R Analysis GazeAnalysis I-VT filter)	Interpolated Distance (R Analysis GazeAnalysis I-VT filter)	Gaze Width
1	527.6065	1	StartSlide	TestImage	200000		StimuliDisplay	Welcome						
2	527.6214						Welcome	1	Welcome	741.5	260.5	744	263.3333333333333	181.364602462985
3	529.0496						Welcome	1	Welcome	755	260.5	753	261.3333333333333	127.57174198043
4	530.9212						Welcome	1	Welcome	762.5	260.5	760.3333333333333	599.052276611328	93.7542725505294
5	532.991						Welcome	1	Welcome	766.5	259.5	766	599.1112976027422	68.7901680214479
6	534.0453						Welcome	1	Welcome	769	261	767.6666666666667	599.121673583984	48.8917159695434
7	536.1879						Welcome	1	Welcome	767.5	259.5	760.6666666666667	599.11145019531	34.3204885701252
8	538.1138						Welcome	1	Welcome	765	261.5	764.3333333333333	599.13647460938	20.2466307993318
9	539.0081						Welcome	1	Welcome	760.5	259	761.3333333333333	599.106262207031	10.7840156916406
10	540.9944						Welcome	1	Welcome	758.5	261.5	758.6666666666667	599.140680351156	0.12151529579585
11	542.0111						Welcome	1	Welcome	757	260.5	757.1666666666667	599.13402320113	6.0058960520072
12	544.0741						Welcome	1	Welcome	756	262	761.1666666666667	599.137756547656	9.57319424632002
13	545.9147						Welcome	1	Welcome	756.5	260.5	758.1666666666667	599.142963867919	10.3594405192582

Respondents (1/5) | Sensor data (25/185)

Metric	Detailed description	Unit
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Gaze X	Gaze data X-coordinate (relative to top-left corner of the screen). Average of the X-coordinates of the left and right eye.	Pixels (px)
Gaze Y	Gaze data Y-coordinate (relative to top-left corner of the screen). Average of the Y-coordinates of the left and right eye.	Pixels (px)
Interpolated Gaze X	Gaze data X-coordinate (relative to top-left corner of the screen) with missing coordinates interpolated. Average of the X-coordinates of the left and right eye. Linear interpolation from the last sample before the gap until the first sample after the gap.	Pixels (px)
Interpolated Gaze Y	Gaze data Y-coordinate (relative to top-left corner of the screen) with missing coordinates interpolated. Average of the Y-coordinates of the left and right eye. Linear interpolation from the last sample before the gap until the first sample after the gap.	Pixels (px)
Interpolated Distance	Estimated distance between the eye-tracker and the eyes, with missing values interpolated. Average of the eye tracker's distance to the left and right eye. Linear interpolation from the last sample before the gap until the first sample after the gap.	Millimeters (mm)
Gaze Velocity	Angular velocity of the gaze at the current sample point, i.e. how fast the eyes are moving at this point in time. Computation is based on a time window around the current sample. Distance between the first and last sample of this window is divided by the the time lapse between them.	Degrees per second (°/s)

Gaze Acceleration	Angular acceleration of the gaze at the current sample point, i.e. how much the eyes increased in speed at the current sample. Computed as the difference in velocity between the current and previous sample.	Degrees per second squared ($^{\circ}/s^2$)
Fixation Index	Fixation number, counting from start of the recording	
Fixation Index by Stimulus	Fixation number, counting from start of the stimulus. If first or last fixation is partly taking place during the previous or subsequent stimulus, it is excluded.	
Fixation X	X-coordinate of the fixation centroid (relative to top-left corner of the screen)	Pixels (px)
Fixation Y	Y-coordinate of the fixation centroid (relative to top-left corner of the screen)	Pixels (px)
Fixation Start	Start time of the fixation (relative to recording start), calculated as the average between the timestamp of the first sample of the fixation and the sample preceding it.	Milliseconds (ms)
Fixation End	End time of the fixation (relative to recording start), calculated as the average of the timestamps of the last sample of the fixation and the one succeeding it.	Milliseconds (ms)
Fixation Duration	Duration of the fixation, calculated as the time difference between fixation's start and end times.	Milliseconds (ms)
Fixation Dispersion	Dispersion of the fixation, i.e. how much the fixation's gaze points are spread out. Computed as root mean square of the samples belonging to the fixation.	Degrees ($^{\circ}$)
Saccade Index	Saccade number, counting from start of the recording	
Saccade Index by Stimulus	Saccade number, counting from start of the stimulus. If first or last saccade is partly taking place during the previous or subsequent stimulus, it is excluded.	

Saccade Start	Start time of the saccade (relative to recording start), calculated as the average between the timestamp of the first sample of the saccade and the sample preceding it.	Milliseconds (ms)
Saccade End	End time of the saccade (relative to recording start), calculated as the average of the timestamps of the last sample of the saccade and the one succeeding it.	Milliseconds (ms)
Saccade Duration	Duration of the saccade, calculated as the time difference between saccade's start and end times.	Milliseconds (ms)
Saccade Amplitude	Amplitude of the saccade, i.e. angular distance that the eyes travelled from start point to end point	Degrees (°)
Saccade Peak Velocity	Peak velocity of the saccade, i.e. the maximal speed of the eyes during this saccade	Degrees per second (°/s)
Saccade Peak Acceleration	Peak acceleration of the saccade, i.e. the maximal increase in speed of the eyes during this saccade	Degrees per second squared (°/s ²)
Saccade Peak Deceleration	Peak deceleration of the saccade, i.e. the maximal decrease in speed of the eyes during this saccade	Degrees per second squared (°/s ²)
Saccade Direction	Direction of the saccade from its start point to end point, indicated as clockwise angles: 0 degrees mean a horizontal saccade from left to right, 90 degrees a vertical saccade from top to bottom.	Degrees (°)

Important note about correct screen parameters:

To compute gaze angles, iMotions has to convert the difference in gaze coordinates (in pixels on the x- and y-axes) into gaze angles. For this, it is using the width and height from the display information populated in the study settings. For this computation to be correct, it is absolutely crucial that the parameters reflect the true size of your computer screen and its resolution, or size of the scene camera's field of view and its resolution, respectively, that you used during data collection.

For screen based eye trackers, the algorithm uses the distance between eyes and screen as it was recorded by the eye tracker in order to convert the distance on the screen into a gaze angle. The width and height of your computer screen need to be manually measured with a ruler, and filled in accordingly.

For mobile eye tracking glasses and VR headsets, the algorithm uses the fixed field of view parameters (width, height, and distance) as they are entered in study settings. Newer versions of iMotions auto-populate these fields with the manufacturer's information; if you are using an older version, you need to enter the field of view parameters manually:

- Tobii Glasses 2: Distance 100 cm, width 173.86 cm, height 97.55 cm with resolution 1920 x 1080
- Argus Science ET3Vision: Distance 100 cm, width 207.11 cm, height 97.55 cm with resolution 1280 x 720
- Pupil Invisible: Distance 100 cm, width 200 cm, height 200 cm with resolution 1088 x 1080
- Varjo: Distance 100 cm, width 184 cm, height 170 cm with the automatically populated resolution. Please note that average values are being used for VR headsets which can deviate slightly from the actual field of view resulting from the individual configuration of the headset.
- Pupil Core: The parameters depend on the settings that you chose in the Pupil app. Therefore, they always need to be updated manually:
 - Wide Angle Lens 1080p: Distance 100 cm, width 535 cm, height 177 cm
 - Wide Angle Lens 720p: Distance 100 cm, width 234 cm, height 100 cm
 - Wide Angle Lens 480p: Distance 100 cm, width 238 cm, height 151 cm
 - Narrow Angle Lens 1080p: Distance 100 cm, width 193 cm, height 102 cm
 - Narrow Angle Lens 720p: Distance 100 cm, width 123 cm, height 67 cm
 - Narrow Angle Lens 480p: Distance 100 cm, width 77 cm, height

57 cm

STUDY SETTINGS SENSORS GAZE ANALYSIS LIVE MARKERS EXPORT RECORDED DATA LATENCY SETTINGS

Name

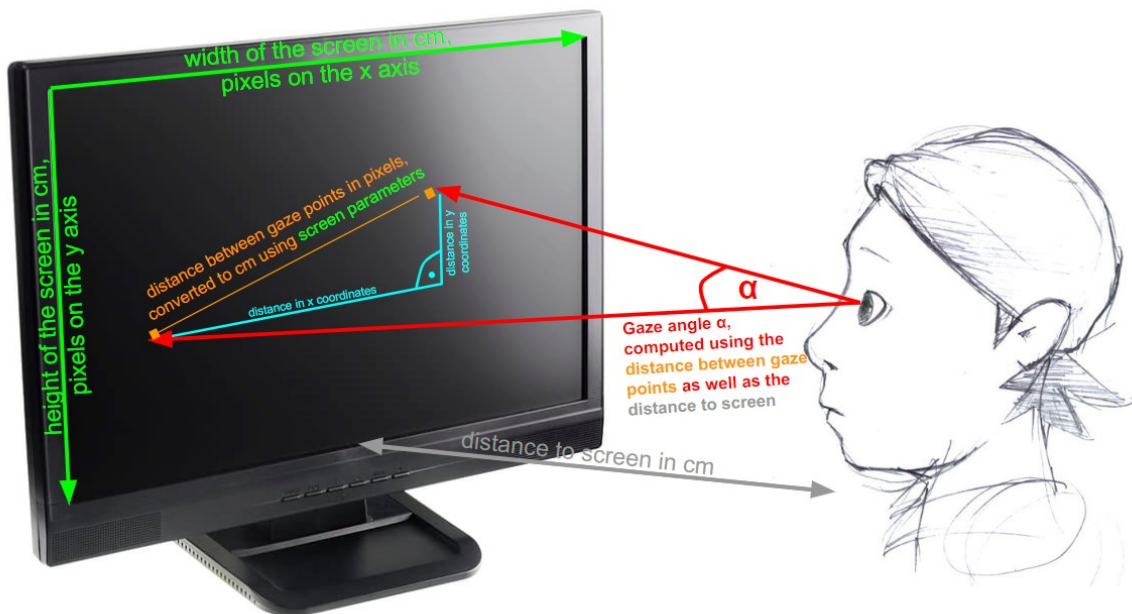
Resolution 1920x1080 ☐ Custom Resolution

Display Size Distance cm

Height cm Width cm

Stimuli Order ☒ Randomize stimuli except for the locked stimuli
☐ Order stimuli as displayed in the stimuli navigator
☐ Load stimuli order from a test plan

Stimuli Blocks ☒ Enable ☒ Auto select block to run



Further readings:

Holmqvist, K., Nyström, N., Andersson, R., Dewhurst, R., Jarodzka, H., & Van de Weijer, J. (Eds.) (2011). Eye tracking: a comprehensive guide to methods and measures, Oxford, UK: Oxford University Press.

Komogortsev, O. V., Gobert, D. V., Jayarathna, S., Koh, D. H., & Gowda, S. M. (2010). Standardization of automated analyses of oculomotor fixation and saccadic behaviors. *Biomedical Engineering, IEEE Transactions on*, 57(11), 2635-2645.

